

Python Code

```
1  from fastapi import FastAPI, Request, Response, HTTPException
2  from fastapi.middleware.cors import CORSMiddleware
3  import httpx
4  import uvicorn
5
6  app = FastAPI(title="AKMP API Gateway")
7
8  # Enable CORS for frontend
9  app.add_middleware(
10     CORSMiddleware,
11     allow_origins=["*"],
12     allow_credentials=True,
13     allow_methods=["*"],
14     allow_headers=["*"],
15 )
16
17 SPRING_BOOT_URL = "http://localhost:8080/api"
18
19 @app.get("/")
20 def read_root():
21     return {"message": "Welcome to AKMP API Gateway"}
22
23 async def forward_request(request: Request, path: str, target_url: str):
24     url = f"{target_url}/{path}"
25
26     # Extract query params
27     params = dict(request.query_params)
28
29     # Extract headers (filtering out some hop-by-hop headers if necessary)
30     headers = dict(request.headers)
31     headers.pop("host", None)
32
33     async with httpx.AsyncClient() as client:
34         try:
35             # Handle request body for methods that support it
36             body = await request.body()
37             response = await client.request(
38                 method=request.method,
39                 url=url,
40                 headers=headers,
41                 params=params,
42                 content=body
43             )
44             return Response(content=response.content,
45                             status_code=response.status_code, headers=dict(response.headers))
46         except httpx.RequestError as exc:
47             raise HTTPException(status_code=503, detail=f"Service unavailable: {str(exc)}")
```

```
48 @app.api_route("/api/v1/auth/{path:path}", methods=["GET", "POST", "PUT", "DELETE",  
49 "PATCH"])  
49 async def proxy_auth(request: Request, path: str):  
50     # Route to Spring Boot Auth controller  
51     return await forward_request(request, f"auth/{path}", SPRING_BOOT_URL)  
52  
53 @app.api_route("/api/v1/sql/{path:path}", methods=["GET", "POST", "PUT", "DELETE",  
54 "PATCH"])  
54 async def proxy_sql(request: Request, path: str):  
55     # Route to Spring Boot standard controllers  
56     return await forward_request(request, path, SPRING_BOOT_URL)  
57  
58 if __name__ == "__main__":  
59     uvicorn.run(app, host="0.0.0.0", port=8000)  
60
```